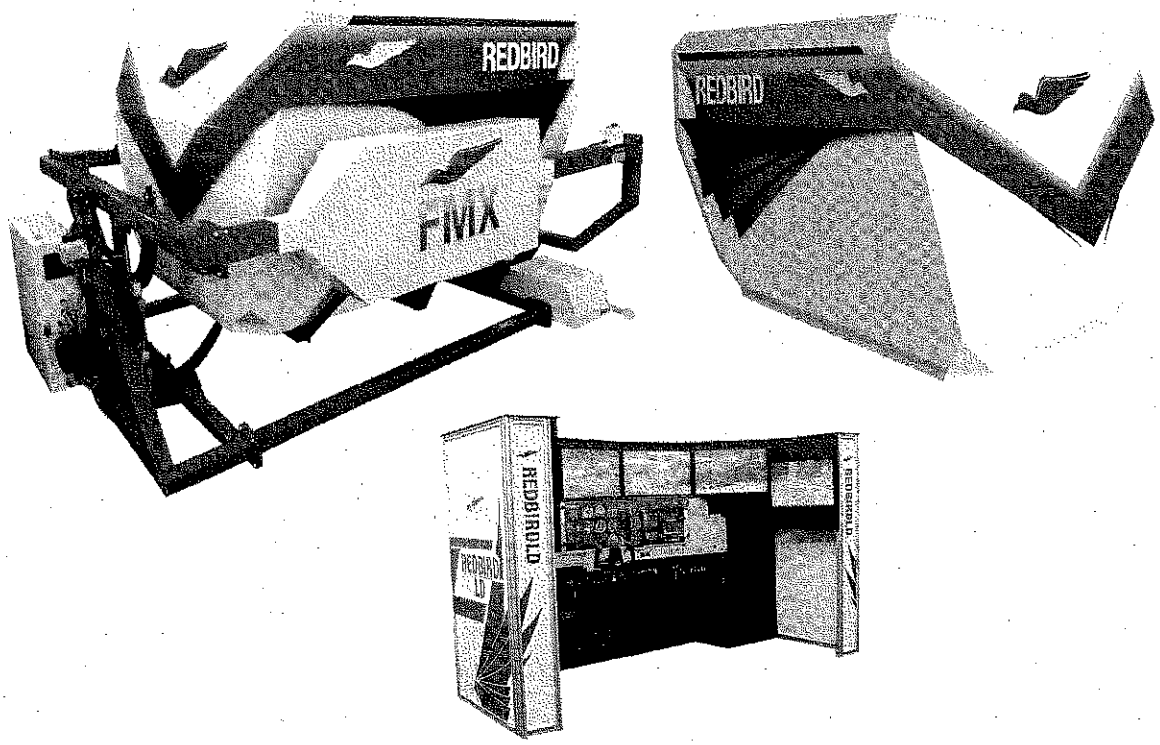
Redbird Flight Simulations
Qualification and Approval Guide (QAG)

LD 1000, SD 1000, and FMX 1000 Advanced Aviation Training Devices
Amended Qualification and Approval Guide
VERSION 4.3



FAA
Aviation Safety

Flight Standards Service AFS-800
General Aviation and
Commercial Division

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APPROVED QAG

Date 6/27/2012

Signature

Section 1: List of Configurations

Currently Approved:

1. **Configuration: Generic Complex Single Engine - *Renamed RB10-S1* –** Single Engine, Land, Retractable Gear, Constant Speed Propeller, High Wing

2. **Configuration: C172-XX-S001 - *Renamed C172-S1* -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, High Wing

3. **Configuration: C172-XX-G001 - *Renamed C172-G1* -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, High Wing

4. **Configuration: C182-XX-S001 - *Renamed C182-S1* -** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

5. **Configuration: C182-XX-G001 - *Renamed C182-G1* -** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

6. **Configuration: PA28A-XX-S001 - *Renamed PA28-60-S2* -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

7. **Configuration: PA28R-XX-S001 - *Renamed PA28-2R-S1* -** Single Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

8. **Configuration: SR20-XX-G001 - *Renamed SR20-G1* -** Single Engine, Land, Fixed Gear, Constant Speed Propeller, Low Wing

9. **Configuration: SR22-XX-G001 - *Renamed SR22-G1* -** Single Engine, Land, Fixed Gear, Constant Speed Propeller, Low Wing

10. **Configuration: BE58-XX-S001 - *Renamed BE58-S1* -** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

11. **Configuration: PA34-XX-S001 - *Renamed PA34-S1* -** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

12. **Configuration: PA44-XX-S001 - *Renamed PA44-S1* -** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

Currently Approved:

13. **Configuration: C172-FC-G003 - *Renamed C172-G2* -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, High Wing

14. **Configuration: C182-FC-G003 - *Renamed C182-G2* -** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

15. **Configuration: PA28A-XX-G001 - *Renamed PA28-60-G1* -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

16. **Configuration: PA28B-XX-S001 - *Renamed PA28-80-S1* -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

17. **Configuration: PA28R-XX-G001 - *Renamed PA28-2R-G1* -** Single Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

18. **Configuration: DA20-XX-S007 - *Renamed DA20-S1* -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

19. **Configuration: DA40-FC-G003 - *Renamed DA40-G1* -** Single Engine, Land, Fixed Gear, Constant Speed Propeller, Low Wing

20. **Configuration: SR22-FC-G004 - *Renamed SR22-G2* -** Single Engine, Land, Fixed Gear, Constant Speed Propeller, Low Wing

21. **Configuration: BE55-XX-S008 - *Renamed BE55-S1* -** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

22. **Configuration: BE58-FC-G003 - *Renamed BE58-G1* -** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

23. **Configuration: BE76-XX-S006 - *Renamed BE76-S1* -** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

24. **Configuration: PA30-XX-S009 - *Renamed PA30-S1* -** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

25. **Configuration: GA7-XX-S009 - *Renamed GA7-S1* -** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

26. **Configuration: DA42-FC-G003 - *Renamed DA42-G1* -** Twin Engine, Land, Retractable Gear, Constant Speed Propeller, Low Wing

Additional Configurations Included in this Version (4.3):

1. **Configuration: C172-S2 -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, High Wing

2. **Configuration: C182-S2 -** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

3. **Configuration: C206-S1-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

4. **Configuration: C206-G1-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

5. **Configuration: C206-G2-** Single Engine, Land, Fixed Gear, Constant Speed Propeller, High Wing

6. **Configuration: BE36-S1-** Single Engine, Land, Retractable Gear, Constant Pitch Propeller, Low Wing

7. **Configuration: BE36-G1-** Single Engine, Land, Retractable Gear, Constant Pitch Propeller, Low Wing

8. **Configuration: LXL2-S1 -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

9. **Configuration: LXL2-S2 -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, Low Wing

Statements of Compatibility of Software and Hardware
AC61-136, Appendix 2, 4. b.:

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Microsoft Windows XP or 7 (Operating System)
This is to certify that Microsoft Corporation, the owner and developer of the Windows 7 operating system, has evaluated that their operating system works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

Microsoft ESP or FSX (Simulation Engine)
This is to certify that Microsoft Corporation, the owner and developer of the ESP and FSX Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

Lockheed Martin Prepar3D (Simulation Engine)
This is to certify that Lockheed Martin, the owner and developer of the Prepar3D Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

Mindstar Aviation Redbird Virtual Instrument Suite
This is to certify that Mindstar Aviation, the owner and developer of the Redbird Virtual Instrument Suite, including the Redbird 1000, Redbird 430/530, Redbird KLN94, Redbird 140 Autopilot, Redbird 55 Autopilot, Redbird Radios, Redbird Entegra and Redbird Perspective and Redbird Flight Simulations, the hardware manufacturer and component integrator, have demonstrated that the Redbird Virtual Instrument Suite software package is fully compatible with the Redbird Flight Simulations LD, SD, FMX 1000 and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations.

Statements of Compatibility of Software and Hardware
AC61-136, Appendix 2, 4. b. (continued):

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Flight 1 Compass

"This is to certify that Flight 1, the owner and developer of the Compass and Redbird Flight Simulations, the hardware manufacturer and component integrator, have demonstrated that the Compass software is fully compatible with the Redbird Flight Simulations LD, SD, FMX 1000 and all configurations of that model. Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations."

Section 2: Description of Hardware and Software Components

All Configurations Include:

Qty	Type	Manufacturer	Name	Description/Function	Configuration
1	Software	Microsoft	Windows XP or 7	Operating system. (depending on serial number)	All
1	Software	Microsoft	ESP or FSX	Simulation engine. (depending on serial number)	All
1	Software	Lockheed Martin	Prepar3D	Simulation engine. (depending on serial number)	All
1	Software	Redbird Flight Simulations	RB Sim	Simulation control and component integration.	All
1	Software	Redbird Flight Simulations	FMX Control	Motion system control and component integration.	FMX only
1	Software	Redbird Flight Simulations	Instructors Station	Environmental, location and failure controls with map, track and glideslope display.	All
1	Software	Mindstar Aviation	Redbird 1000	Virtual flight instruments, GPS, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	See Configuration
1	Software	Mindstar Aviation	Redbird 430/ 530	Virtual GPS and radios	See Configuration
1	Software	Mindstar Aviation	Redbird KLN94	Virtual GPS	See Configuration
1	Software	Mindstar Aviation	Redbird EFD1000	Virtual EFD	See Configuration
1	Software	Mindstar Aviation	Redbird Entegra	Virtual flight instruments, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	See Configuration
1	Software	Mindstar Aviation	Redbird Autopilot	Virtual Autopilot (Redbird 140, 55, or 700)	See Configuration
1	Software	Mindstar Aviation	Redbird Radios	Virtual Radios (KX155, SL40)	See Configuration
1	Software	Mindstar Aviation	Miscellaneous Gauges	Virtual Miscellaneous Gauges	See Configuration
1	Software	Flight 1	Compass	Virtual Compass	All
1	Software	Redbird Flight Simulations	Analog Gauges	Virtual Airspeed, Attitude, Altimeter, Turn and Bank, Heading, HSI, VSI, RMI, CDI, and ADF Gauges , Radios	See Configuration
1	Software	Redbird Flight Simulations	Miscellaneous Gauges	Virtual Miscellaneous Gauges	All
1	Hardware	Redbird Flight Simulations	Simulation Computer	Host computer for flight simulation engine, simulation control software, airplane systems and instruments.	All
1	Hardware	Redbird Flight Simulations	Motion Control Computer	Host computer for FMX Control.	FMX only
1	Hardware	Industry Standard	Instructors Station Computer	Host computer for Instructors Station.	All
1	Hardware	Redbird Flight Simulations	Motion Platform	Gimbaled, steel motion platform with movement in pitch, roll and yaw. Includes all motors, sensors and safety controls.	FMX only- See Figure 5.9
1	Hardware	Redbird Flight Simulations	Cockpit Enclosure	Cockpit enclosure to exclude distractions.	All- See Figure 5.7, 5.8,5.9
6	Hardware	22” Industry Standard	LCD	Flat Panel displays for exterior views.	All
2	Hardware	19” Industry Standard	LCD	Flat Panel displays for virtual instruments.	All
1	Hardware	Redbird Flight Simulations	Yoke, Side Yoke, Center Stick	Pitch and roll controller with switches and buttons for airplane systems operation. As required for each configuration.	All- See Configuration
1	Hardware	Redbird Flight Simulations	Throttle Quadrant	As required for each configuration.	All- See Configuration

Qty	Type	Manufacturer	Name	Description/Function	Configuration
2	Hardware	Redbird Flight Simulations	Rudder Pedals	Rudder control pedals with toe brakes.	All
1	Hardware	Redbird Flight Simulations	Switch Panel	Lower switch panel with airplane configuration and systems controls.	All
1	Hardware	Redbird Flight Simulations	Instrument Controls Overlay	Flight instruments, radios, airplane configuration and systems controls as required for each configuration.	All- See Configuration

Approved Individual Configurations Include:

RB10-S1: Approved Configuration
Yoke: Standard yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 3.2
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530,Redbird 140 Autopilot. See Figure— 1.1

C172-S1: Approved Configuration
Yoke: Standard yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Vernier. See Figure— 3.1
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.2

C172-G1: Approved Configuration
Yoke: Standard yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Vernier. See Figure— 3.1
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Redbird 140 Autopilot. See Figure—1.3

C182-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 3.2
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530,Redbird 140 Autopilot. See Figure— 1.5

C182-G1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 3.2
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Redbird 140 Autopilot. See Figure—1.6

PA28-60-S2: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 3.3
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.8

PA28-2R-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex “Boat Style”. See Figure— 3.4
Virtual Instrument Controls Overlay: Traditional Round Gauges, Dual Redbird 430, Redbird 55 Autopilot. See Figure— 1.11

SR20-G1: Approved Configuration
Yoke: Side Yoke. See Figure— 2.2
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 3.3
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot. See Figure— 1.15

SR22-G1: Approved Configuration
Yoke: Side Yoke. See Figure— 2.2
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 3.3
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot. See Figure— 1.16

BE58-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 3.5
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.19

PA34-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 3.5
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.23

PA44-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 3.5
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.24

C172-G2: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Vernier. See Figure— 3.1
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 1.4

C182-G2: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 3.2
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 1.7

PA28-60-G1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 3.3
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot. See Figure— 1.9

PA28-80-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 3.3
Virtual Instrument Controls Overlay: Traditional Round Gauges, Dual Redbird 430, Redbird 55 Autopilot. See Figure— 1.10

PA28-2R-G1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex “Boat Style”. See Figure— 3.4
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot. See Figure—1.12

DA20-S1: Approved Configuration
Yoke: Center Stick. See Figure— 2.3
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 3.3
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430, Redbird 140 Autopilot. See Figure— 1.13

DA40-G1: Approved Configuration
Yoke: Center Yoke. See Figure— 2.3
Throttle Quadrant: Single Engine Complex “Boat Style”. See Figure— 3.4
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 1.14

SR22-G2: Approved Configuration
Yoke: Side Yoke. See Figure— 2.2
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 3.3
Virtual Instrument Controls Overlay: Glass Cockpit Redbird Perspective PFD & MFD with Redbird FMS Keypad and Redbird Perspective Autopilot. See Figure— 1.17

BE55-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Twin Engine Reverse “Boat Style”. See Figure— 3.6
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.18

BE58-G1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 3.5
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 1.20

BE76-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 3.5
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.21

PA30-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 3.5
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.22

DA42-G1: Approved Configuration
Yoke: Center Stick. See Figure— 2.3
Throttle Quadrant: Twin Engine FADEC “Boat Style”. See Figure— 3.7
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 1.25

GA7-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Twin Engine “Boat Style”. See Figure— 3.5
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.26

Additional configurations included in this version:

C172-S2: Approved Configuration
Yoke: Standard yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Vernier. See Figure— 3.1
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird KLN94, Redbird 140 Autopilot. See Figure— 1.27

C182-S2: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 3.2
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird KLN94, Redbird 140 Autopilot. See Figure— 1.28

C206-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 3.2
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530,Redbird 140 Autopilot. See Figure— 1.29

C206-G1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 3.2
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Redbird 140 Autopilot. See Figure— 1.30

C206-G2: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex Vernier. See Figure— 3.2
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 1.31

BE36-S1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex “Boat Style”. See Figure— 3.4
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 430/530, Redbird 140 Autopilot. See Figure— 1.32

BE36-G1: Approved Configuration
Yoke: Standard Yoke. See Figure— 2.1
Throttle Quadrant: Single Engine Complex “Boat Style”. See Figure— 3.4
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 1.33

LXL2-S1: Approved Configuration
Yoke: Center Stick. See Figure— 2.3
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 3.3
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 530/430, Redbird 55 Autopilot. See Figure— 1.34

LXL2-S2: Approved Configuration
Yoke: Center Stick. See Figure— 2.3
Throttle Quadrant: Single Engine “Boat Style”. See Figure— 3.3
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird EFD1000, Redbird 430, Redbird SL40, Redbird 55 Autopilot. See Figure— 1.35

Section 3: Statements of Compliance

All Configurations, as noted, meet AC 61-136, Appendix 2, Paragraph 8 BATD Design Criteria:

a. Controls:

(1) Meets Requirements: Physical flight and aircraft system controls are recognizable as to their function and how they are manipulated solely from their appearance. There is no keyboard, mouse, or gaming joystick inside the LD/SD/FMX 1000 Cockpit. See Figures— 5.5, 5.6

(2) Meets Requirements: Virtual controls are limited to the initialization of the device and instructor input. Avionics are virtually displayed but are operated with physical buttons and knobs that closely represent the configuration of the aircraft. See Figures— 4.1, 4.2, 5.5, 5.6

(3) Meets Requirements: No keyboard or mouse is used to set or position any feature of the device. The required and additional equipment is operated in the same manner in which it would be operated in the represent aircraft.

(4) Meets Requirements: The arrangement, appearance, and operation of controls, instruments, and switches closely represent the aircraft, including: master/battery, magnetos, alternators, fuel boost pumps, avionics master, pitot heat, and required lights. See Figures— 4.1, 4.2, 5.5, 5.6

(5) Meets Requirements: During start-up of the device the PC configuration and file inventory is examined to ensure only tested software is installed. In the event additional files are found, an initialization error will be displayed and logging of the flight training data to the student key will not be allowed.

b. Control Requirements:

(1)(a): Meets Requirements: See individual configuration.
(1)(b): Meets Requirements: See Figure— 5.3
(1)(c): Meets Requirements: See Figure— 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7
(1)(d): Meets Requirements: See individual configuration.
(1)(e): Meets Requirements: See individual configuration.

c. Control Inputs:

(1) Meets Requirements: The transport delay for all control inputs is 300 milliseconds or less. The required transport delay is tested at system start-up by internal start/stop timers. Redbird Flight Simulations is the manufacturer of all control inputs and uses USB 2.0 which exceeds this requirement.

(2) Meets Requirements: Each control input is tested at training session start-up. If any device fails to meet the requirements of Appendix 2, Paragraph 8, Section c, item (1) a warning message is displayed and the device will not start a training session.

d. Display Requirements:

(1) Meets Requirements: See individual configuration.
(2) Meets Requirements: All instrument displays included in each individual configuration are visible during all flight operations. The displays are updated at rate that insures that the image of the instrument does not appear to: be out of focus, illegible, jump, step, jagged, nor is it distracting.
(3) Meets Requirements: Display updates exceed 60Hz. All replicated instruments display changes equal or greater than the values required by this item.
(4) Meets Requirements: Displays reflect the dynamic behavior of an actual aircraft in flight.

e. Flight Dynamics Requirements:

(1) Meets Requirements: All simulated aircraft flight dynamics are comparable to the way an actual aircraft performs and handles.
(2) Meets Requirements: All simulated aircraft performance parameters are comparable to the aircraft being represented.
(3) Meets Requirements: Vertical lift component changes as a function of bank, comparable to the aircraft being represented.
(4) Meets Requirements: Changes in aircraft configuration are accompanied by changes in flight dynamics, comparable to the aircraft being represented.
(5) Meets Requirements: The handling and performance qualities of the simulated aircraft are affected by the presence and intensity of wind and turbulence in manner comparable to the aircraft being represented.

f. Instructional Management Requirements:

(1) Meets Requirements: The instructor has the ability to pause the system at any time using either the buttons on the instructor's console or the buttons on the instrument panel. See Figure— 5.2, 5.1
(2) Meets Requirements: The instructor as the ability to manipulate the parameters listed in this item independently of the simulation.
(3) Meets Requirements: The system records both the horizontal and vertical track of the aircraft movement and this can be view in real-time or at a later time.
(4) Meets Requirements: The instructor is able to disable instruments prior to or during a training session and simulate failure of the instruments without stopping or freezing the simulation to affect the failure.
(5) Meets Requirements: The device includes a navigational data base that covers all of North America and is based on procedures as published per 14 CFR part 97.

All configurations, as noted meet AC 61-136, Appendix 3, Paragraph 3 AATD Design Criteria:

a.

(1) Meets Requirements: The device enclosure and cockpit design and instrument panel arrangement represent a generic aircraft commonly used by flight training operations. See Figure— 5.5, 5.6, 5.7, 5.8, 5.9
(2) Meets Requirements: Cockpit knobs/system controls/switches/switch panels are realistic in size and design appropriate to each intended function, in the proper position and distance from the pilot's seated position, and representative of the category and class of the aircraft being represented. See individual configuration.

(3) Meets Requirements: Primary flight and navigation instruments are life size and exhibit neither stepping nor excessive transport delay. They are arranged so as to observe trends and provide a realistic scan pattern. See individual configuration.
(4) Meets Requirements: See individual configuration.
(5) Meets Requirements: See individual configuration.
(6) Meets Requirements: See individual configuration.
(7) Meets Requirements: Manual pitch trim is provided via a trim wheel on all configurations. Individual configurations may have electric trim as well. See individual configuration. See Figure— 4.3
(8) Meets Requirements: All visual cues for both day and night, VFR and IFR are provided. See Figure— 5.4
(9) Meets Requirements: Instructors station is run from a “tablet” style or laptop PC from inside or outside the device. The instructor can oversee the pilot’s horizontal and vertical flight profiles in real time and space. The instructor has the ability to complete all functions listed in this item without interrupting the flight. See Figure— 5.2
(10) Cockpit seat is fully adjustable.

Approved Configuration: RB10-S1

This configuration represents a generic, high wing, complex, single engine land, airplane.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.2
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.1

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.1
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels.
See Figure— 1.1

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated.
See Figure— 1.1

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot.
See Figure— 1.1

(5) Redbird 430/530. See Figure— 1.1

(6) Redbird 140 Autopilot. See Figure— 1.1

(7) Manual pitch trim with indicator. See Figures— 4.3, 1.1

Approved Configuration: C172-S1

This configuration represents a single engine land, fixed gear, fixed pitch propeller, high wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.1
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.2

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.2
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item. Gyroscopic rate-of-turn indicator that meets the requirements listed in this item. Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item. Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.2

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.2

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.2

(5) Redbird 430/530. See Figure— 1.2

(6) Redbird 140 Autopilot. See Figure—1. 2

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.2

Approved Configuration: C172-G1

This configuration represents a single engine land, fixed gear, fixed pitch propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and a Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.1
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.
This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.3

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.3
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item. Gyroscopic rate-of-turn indicator that meets the requirements listed in this item. Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item. Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.3

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.3

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Redbird 140 Autopilot. See Figure— 1.3

(5) Glass Cockpit Redbird 1000 PFD & MFD . See Figure—1.3

(6) Redbird 140 Autopilot. See Figure— 1.3

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.3

Approved Configuration: C182-S1

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.2
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.5

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.5
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.5

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.5

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.5

(5) Redbird 430/530. See Figure— 1.5

(6) Redbird 140 Autopilot. See Figure— 1.5

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.5

Approved Configuration: C182-G1

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and a Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.2
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.6

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.6
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.6

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.6

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Redbird 140 Autopilot. See Figure— 1.6

(5) Glass Cockpit Redbird 1000 PFD & MFD . See Figure— 1.6

(6) Redbird 140 Autopilot. See Figure— 1.6

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.6

Approved Configuration: PA28-60-S2

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.3
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.8

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.8
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.8

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.8

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.8

(5) Redbird 430/530. See Figure— 1.8

(6) Redbird 140 Autopilot. See Figure— 1.8

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.8

Approved Configuration: PA28-2R-S1

This configuration represents a single engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird Dual 430, and Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.4
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.11

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.11
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item(Dual Redbird 430).
Navigation radio that meets the requirements listed in this item (Dual Redbird 430).

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure—1.11

(3) Life sized traditional “round” gauges and Redbird Dual 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.11

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.11

(5) Redbird 430/530. See Figure— 1.11

(6) Redbird 55 Autopilot. See Figure— 1.11

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.11

Approved Configuration: SR20-G1

This configuration represents a single engine land, fixed gear, constant speed propeller, low wing, airplane with Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering side yoke. See Figure— 2.2
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.3
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.15

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird Entegra PFD & MFD with standby instruments. See Figure-1.15
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Dual Redbird 430)
Navigation radio that meets the requirements listed in this item. (Dual Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird Entegra PFD & MFD with standby instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.15

(3) Life sized Glass Cockpit Redbird Entegra PFD & MFD and dual Redbird 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.15

(4) Dual Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 1.15

(5) Dual Redbird 430 . See Figure— 1.15

(6) Redbird 55 Autopilot. See Figure— 1.15

(7) Manual and electric pitch trim with indicator. See Figure— 4.3, 2.2, 1.15

Approved Configuration: SR22-G1

This configuration represents a single engine land, fixed gear, constant speed propeller, low wing, airplane with Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering side yoke. See Figure— 2.2
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.3
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.16

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird Entegra PFD & MFD with standby instruments. See Figure- 1.16
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Dual Redbird 430)
Navigation radio that meets the requirements listed in this item. (Dual Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird Entegra PFD & MFD with standby instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.16

(3) Life sized Glass Cockpit Redbird Entegra PFD & MFD and dual Redbird 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.16

(4) Dual Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 1.16

(5) Dual Redbird 430. See Figure— 1.16

(6) Redbird 55 Autopilot. See Figure— 1.16

(7) Manual and electric pitch trim with indicator. See Figure— 4.3, 2.2, 1.16

Approved Configuration: BE58-S1

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 3.5

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 4.1, 4.2, 1.19

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.19
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.19

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.19

(4) Redbird 430/530, Transponder, Audio Panel, and Redbird 140 Autopilot. See Figure— 1.19

(5) Redbird 430/530. See Figure— 1.19

(6) Redbird 140 Autopilot. See Figure— 1.19

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.19

Approved Configuration: PA34-S1

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 3.5

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 4.1, 4.2, 1.23

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.23
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.23

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.23

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.23

(5) Redbird 430/530. See Figure— 1.23

(6) Redbird 140 Autopilot. See Figure—1. 23

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.23

Approved Configuration: PA44-S1

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure—3.5

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.
This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 4.1, 4.2, 1.24

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.24
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.24

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.24

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.24

(5) Redbird 430/530. See Figure— 1.24

(6) Redbird 140 Autopilot. See Figure— 1.24

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.24

Approved Configuration: C172-G2

This configuration represents a single engine land, fixed gear, fixed pitch propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.1
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.4

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.4
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.4

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.4

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 1.4

(5) Glass Cockpit Redbird 1000 PFD & MFD . See Figure— 1.4

(6) Integrated Redbird 700 Autopilot. See Figure— 1.4

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.4

Approved Configuration: C182-G2

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.2
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.7

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.7
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass. Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.7

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.7

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 1.7

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 1.7

(6) Integrated Redbird 700 Autopilot. See Figure— 1.7

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.7

Approved Configuration: PA28-60-G1

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.3
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.9

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird Entegra PFD & MFD with standby instruments. See Figure— 1.9
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Dual Redbird 430)
Navigation radio that meets the requirements listed in this item. (Dual Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird Entegra PFD & MFD representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.9

(3) Life sized Glass Cockpit Redbird Entegra PFD & MFD and dual Redbird 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.9

(4) Dual Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 1.9

(5) Dual Redbird 430. See Figure— 1.9

(6) Redbird 55 Autopilot. See Figure— 1.9

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.9

Approved Configuration: PA28-80-S1

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with traditional "round" gauge instruments, dual Redbird 430, and Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.3
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.10

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.10
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass. Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gage applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item (Dual Redbird 430).
Navigation radio that meets the requirements listed in this item (Dual Redbird 430).

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.10

(3) Life sized traditional “round” gauges and dual Redbird 430GPS and Navigation radios representative of aircraft being simulated See Figure— 1.10

(4) Dual Redbird 430, Transponder, Audio Panel, ADF, and Redbird 55 Autopilot. See Figure— 1.10

(5) Redbird 430/530. See Figure— 1.10

(6) Redbird 55 Autopilot. See Figure— 1.10

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.10

Approved Configuration: PA28-2R-G1

This configuration represents a single engine land, retractable gear, constant speed propeller, low wing, airplane with Glass Cockpit Redbird Entegra PFD & MFD, Dual Redbird 430, Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.4
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Dual Redbird 430).
Clock and timer.
Gear handle that functions
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.12

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird Entegra PFD & MFD with standby instruments. See Figure- 1.12
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Dual Redbird 430)
Navigation radio that meets the requirements listed in this item. (Dual Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird Entegra PFD & MFD representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.12

(3) Life sized Glass Cockpit Redbird Entegra PFD & MFD and dual Redbird 430 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.12

(4) Dual Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 1.12

(5) Dual Redbird 430. See Figure— 1.12

(6) Redbird 55 Autopilot. See Figure— 1.12

(7) Manual trim with indicator. See Figure— 4.3, 1.12

Approved Configuration: DA20-S1

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430, and Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement center stick. See Figure— 2.3
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.3
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.13

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented: Traditional flight instruments. See Figure— 1.13
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430)
Navigation radio that meets the requirements listed in this item. (Redbird 430)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.13

(3) Life sized traditional “round” gages and Redbird 430 GPS and Navigation radio representative of aircraft being simulated See Figure— 1.13

(4) Redbird 430, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 1.13

(5) Redbird 430. See Figure— 1.13

(6) Redbird 55 Autopilot. See Figure— 1.13

(7) Manual and electric pitch trim with indicator. See Figure— 4.3, 1.13

Approved Configuration: DA40-G1

This configuration represents a single engine land, fixed gear, constant speed propeller, low wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and a Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement center stick. See Figure— 2.3
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.4
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.14

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.14
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass. Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.14

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.14

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 1.14

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 1.14

(6) Integrated Redbird 700 Autopilot. See Figure— 1.14

(7) Manual and electric pitch trim with indicator. See Figure— 4.3, 1.14

Approved Configuration: SR22-G2

This configuration represents a single engine land, fixed gear, constant speed propeller, low wing, airplane with a Glass Cockpit Redbird Perspective PFD & MFD with Redbird FMS Keypad and Redbird Perspective Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement side yoke. See Figure— 2.2
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.3
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird Perspective).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.17

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird Perspective PFD & MFD with Redbird FMS Keypad and Redbird Perspective Autopilot with standby instruments. See Figure— 1.17
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio meets the requirements listed in this item. (Redbird Perspective)
Navigation radio that meets the requirements listed in this item. (Redbird Perspective)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.17

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.17

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 1.17

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 1.17

(6) Integrated Redbird 700 Autopilot. See Figure— 1.17

(7) Manual pitch and electric trim with indicator. See Figure— 4.3, 2.2, 1.17

Approved Configuration: BE55-S1

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 3.6

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 4.1, 4.2, 1.18

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.18
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.18

(3) Life sized traditional “round” gages and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.18

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.18

(5) Redbird 430/530. See Figure— 1.18

(6) Redbird 140 Autopilot. See Figure— 1.18

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.18

Approved Configuration: BE58-G1

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 3.5

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 4.1, 4.2, 1.20

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.20
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure—1. 20

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.20

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 1.20

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure—1.20

(6) Integrated Redbird 700 Autopilot. See Figure— 1.20

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.20

Approved Configuration: BE76-S1

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 3.5

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 4.1, 4.2, 1.21

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.21
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.21

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.21

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.21

(5) Redbird 430/530. See Figure— 1.21

(6) Redbird 140 Autopilot. See Figure— 1.21

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.21

Approved Configuration: PA30-S1

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 3.5

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 4.1, 4.2, 1.22

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.22
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.22

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated. See Figure— 1.22

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.22

(5) Redbird 430/530. See Figure— 1.22

(6) Redbird 140 Autopilot. See Figure— 1.22

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.22

Approved Configuration: DA42-G1

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement center stick. See Figure— 2.3

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 3.7

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 4.1, 4.2, 1.25

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.25
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.25

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.25

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 1.25

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 1.25

(6) Integrated Redbird 700 Autopilot. See Figure— 1.25

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.25

Approved Configuration: GA7-S1

This configuration represents a twin engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1

(1)(d) This configuration has twin mixture, twin propeller, and twin throttle controls. See Figure— 3.5

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.

See Figures— 4.1, 4.2, 1.26

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.26
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass. Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.26

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated. See Figure— 1.26

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.26

(5) Redbird 430/530. See Figure— 1.26

(6) Redbird 140 Autopilot. See Figure— 1.26

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.26

Approved Configuration: C172-S2

This configuration represents a single engine land, fixed gear, fixed pitch propeller, high wing, airplane with traditional "round" gauge instruments, Redbird KLN94 GPS and Redbird Radios for Navigation and Communication, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture and throttle controls. See Figure— 3.1
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird Radios).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.27

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.27
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item. Gyroscopic rate-of-turn indicator that meets the requirements listed in this item. Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird Radios)
Navigation radio that meets the requirements listed in this item. (Redbird Radios, Redbird KLN94)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.27

(3) Life sized traditional “round” gages, Redbird KLN94 GPS and Redbird Radios for Navigation and Communication representative of aircraft being simulated See Figure— 1.27

(4) Redbird KLN94, Redbird Radios, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.27

(5) Redbird KLN94. See Figure— 1.27

(6) Redbird 140 Autopilot. See Figure—1. 27

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.27

Approved Configuration: C182-S2

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with traditional "round" gauge instruments, Redbird KLN94 GPS and Redbird Radios for Navigation and Communication, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller and throttle controls. See Figure— 3.2
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird Radios).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.28

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.28
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item. Gyroscopic rate-of-turn indicator that meets the requirements listed in this item. Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird Radios)
Navigation radio that meets the requirements listed in this item. (Redbird Radios, Redbird KLN94)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.28

(3) Life sized traditional “round” gages, Redbird KLN94 GPS and Redbird Radios for Navigation and Communication representative of aircraft being simulated See Figure— 1.28

(4) Redbird KLN94, Redbird Radios, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.28

(5) Redbird KLN94. See Figure— 1.28

(6) Redbird 140 Autopilot. See Figure—1. 28

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.28

Approved Configuration: C206-S1

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.2
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.29

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.29
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.29

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.29

(4) Redbird 430/530, Transponder, Audio Panel, ADF, and Redbird 140 Autopilot. See Figure— 1.29

(5) Redbird 430/530. See Figure— 1.29

(6) Redbird 140 Autopilot. See Figure— 1.29

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.29

Approved Configuration: C206-G1

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and a Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.2
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.30

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.30
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.30

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.30

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Redbird 140 Autopilot. See Figure— 1.30

(5) Glass Cockpit Redbird 1000 PFD & MFD . See Figure— 1.30

(6) Redbird 140 Autopilot. See Figure— 1.30

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.30

Approved Configuration: C206-G2

This configuration represents a single engine land, fixed gear, constant speed propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.2
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that does not function by type of aircraft represented.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.31

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.31
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass. Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure— 1.31

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.31

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 1.31

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure— 1.31

(6) Integrated Redbird 700 Autopilot. See Figure— 1.31

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.31

Approved Configuration: BE36-S1

This configuration represents a single engine land, retractable gear, constant speed propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.4
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.32

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.32
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.32

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.32

(4) Redbird 430/530, Transponder, Audio Panel, and Redbird 140 Autopilot. See Figure— 1.32

(5) Redbird 430/530. See Figure— 1.32

(6) Redbird 140 Autopilot. See Figure— 1.32

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.32

Approved Configuration: BE36-G1

This configuration represents a single engine land, retractable gear, constant speed propeller, low wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.1
(1)(d) This configuration has mixture, propeller, and throttle controls. See Figure— 3.4
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.33

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 1.33
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Glass Cockpit Redbird 1000 PFD & MFD with standby by instruments representative of aircraft being simulated with appropriate knobs/system controls/switches/switch panels. See Figure—1. 33

(3) Life sized Glass Cockpit Redbird 1000 PFD & MFD representative of aircraft being simulated. See Figure— 1.33

(4) Glass Cockpit Redbird 1000 PFD & MFD, audio panel, and Integrated Redbird 700 Autopilot. See Figure— 1.33

(5) Glass Cockpit Redbird 1000 PFD & MFD. See Figure—1.33

(6) Integrated Redbird 700 Autopilot. See Figure— 1.33

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.33

Approved Configuration: LXL2-S1

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430/530, and Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.3
(1)(d) This configuration has throttle and fuel cutoff controls. See Figure— 3.3
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430/530).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.34

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.34
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430/530)
Navigation radio that meets the requirements listed in this item. (Redbird 430/530)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.34

(3) Life sized traditional “round” gauges and Redbird 430/530 GPS and Navigation radios representative of aircraft being simulated See Figure— 1.34

(4) Redbird 430/530, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 1.34

(5) Redbird 430/530. See Figure— 1.34

(6) Redbird 55 Autopilot. See Figure— 1.34

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.34

Approved Configuration: LXL2-S2

This configuration represents a single engine land, fixed gear, fixed pitch propeller, low wing, airplane with traditional "round" gauge instruments, Redbird 430, Redbird EFD1000, Redbird SL40 and Redbird 55 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

b. Control Requirements:

(1)(a) This configuration has a self-centering displacement yoke. See Figure— 2.3
(1)(d) This configuration has throttle and fuel cutoff controls. See Figure— 3.3
(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 430 and Redbird SL40).
Clock and timer.
Gear handle that functions.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 4.1, 4.2, 1.35

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Traditional flight instruments. See Figure— 1.35
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 430 and Redbird SL40)
Navigation radio that meets the requirements listed in this item. (Redbird 430 and Redbird SL40)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a.

(2) Traditional “round” gauges representative of aircraft being simulated with appropriate knobs/ system controls/switches/switch panels. See Figure— 1.35

(3) Life sized traditional “round” gauges, Redbird EFD1000 and Redbird 430 GPS and Navigation radio and Redbird SL40 Navigation radio representative of aircraft being simulated See Figure— 1.35

(4) Redbird 430, Redbird SL40, Transponder, Audio Panel, and Redbird 55 Autopilot. See Figure— 1.35

(5) Redbird 430. See Figure— 1.35

(6) Redbird 55 Autopilot. See Figure— 1.35

(7) Manual pitch trim with indicator. See Figure— 4.3, 1.35

Section 4: Figures

Group 1: Virtual Instrument Controls Overlay

Figure 1.1: RB10-S1



Figure 1.2: C172-S1



Figure 1.3: C172-G1



Figure 1.4: C172-G2



Figure 1.5: C182-S1



Figure 1.6: C182-G1



Figure 1.7: C182-G2



Figure 1.8: PA28-60-S2



Figure 1.9: PA28-60-G1



Figure 1.10: PA28-80-S1



Figure 1.11: PA28-2R-S1



Figure 1.12: PA28-2R-G1



Figure 1.13: DA20-S1



Figure 1.14: DA40-G1



Figure 1.15: SR20-G1



Figure 1.16: SR22-G1



Figure 1.17: SR22-G2



Figure 1.18: BE55-S1



Figure 1.19: BE58-S1



Figure 1.20: BE58-G1



Figure 1.21: BE76-S1



Figure 1.22: PA30-S1



Figure 1.23: PA34-S1



Figure 1.24: PA44-S1



Figure 1.25: DA42-G1



Figure 1.26: GA7-S1



Figure 1.27: C172-S2



Figure 1.28: C182-S2



Figure 1.29: C206-S1



Figure 1.30: C206-G1



Figure 1.31: C206-G2



Figure 1.32: BE36-S1



Figure 1.33: BE36-G1



Figure 1.34: LXL2-S1



Figure 1.35: LXL2-S2



Group 2: Yoke

Figure 2.1: Standard Yoke



Figure 2.2: Side Yoke



Figure 2.3: Center Stick



Group 3: Throttle Quadrant

Figure 3.1: Single Engine Vernier Throttle Quadrant



Figure 3.2: Single Engine Complex Vernier Throttle Quadrant



Figure 3.3: Single Engine "Boat Style" Throttle Quadrant

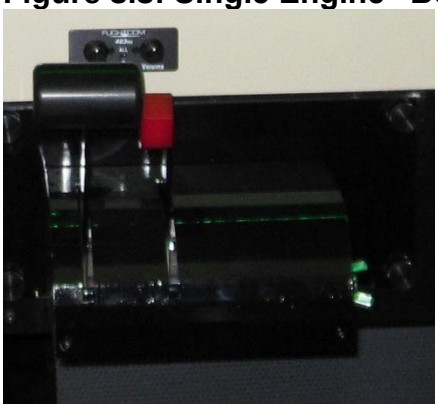


Figure 3.4: Single Engine Complex "Boat Style" Throttle Quadrant

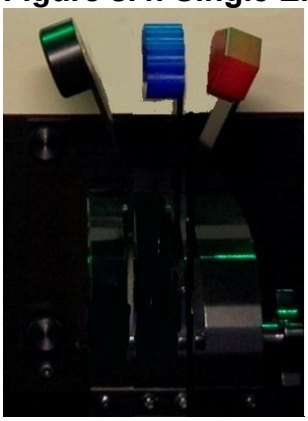


Figure 3.5: Twin Engine “Boat Style” Throttle Quadrant

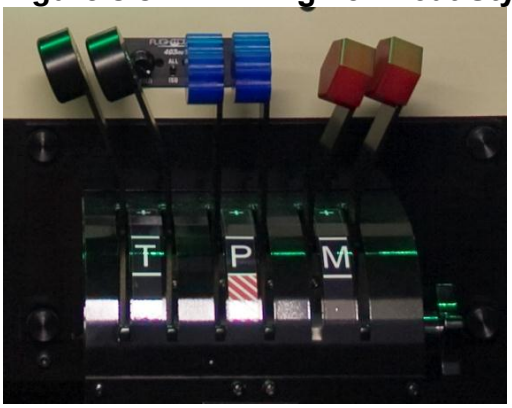


Figure 3.6: Twin Engine Reverse “Boat Style” Throttle Quadrant

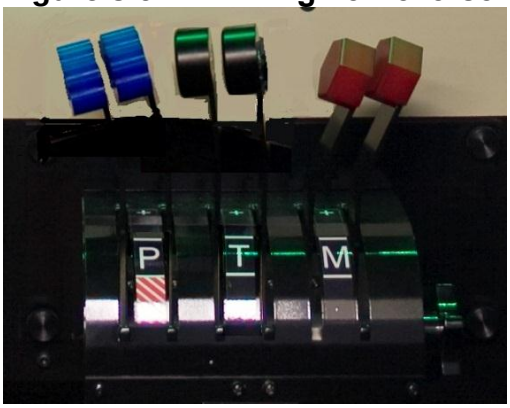
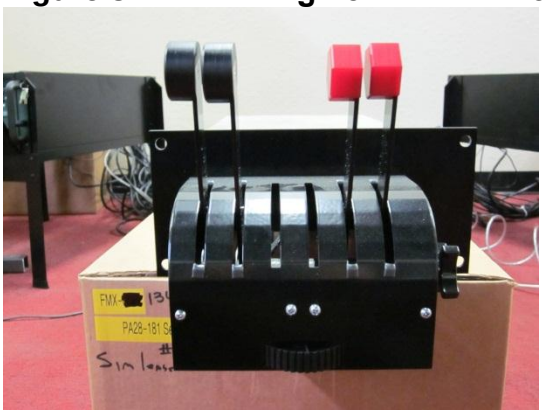


Figure 3.7: Twin Engine FADEC “Boat Style” Throttle Quadrant



Group 4: Switch Panel

Figure 4.1: Right Side of Lower Switch Panel



Figure 4.2: Left Side of Lower Switch Panel

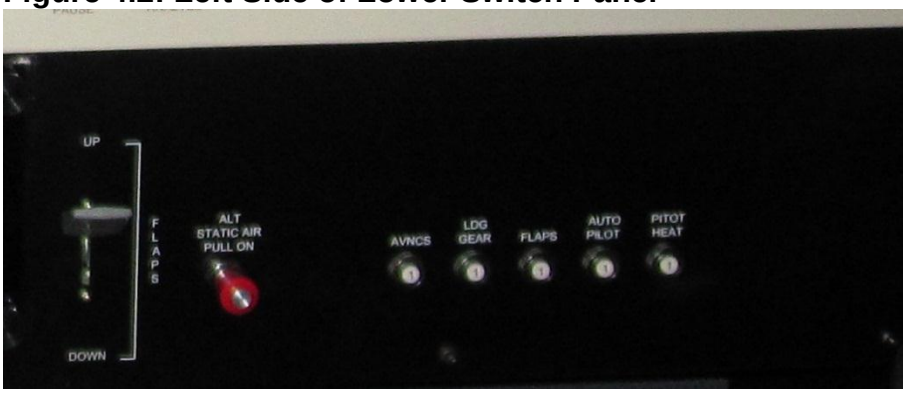


Figure 4.3: Manual Trim Wheel



Group 5: Other

Figure 5.1: Training Session Buttons

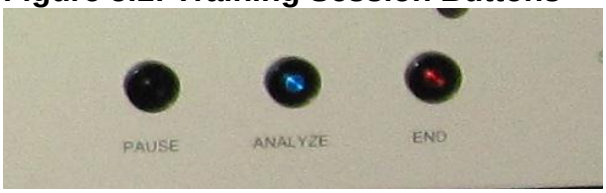


Figure 5.2: Instructors Tablet PC



Figure 5.3: Rudder Pedals

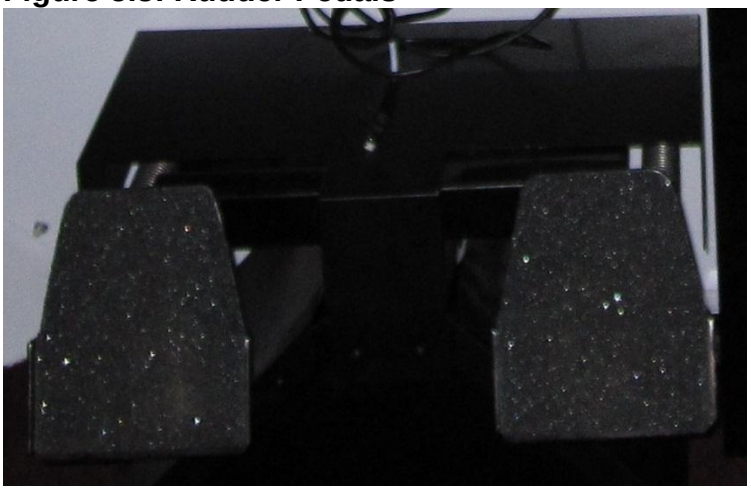


Figure 5.4: Visual Displays



Figure 5.5: Assembled Cockpit Configuration



Figure 5.6: Assembled Cockpit Configuration



Figure 5.7: External Shroud LD



Figure 5.8: External Shroud SD



Figure 5.9: External Shroud FMX

